Lost in Translation, Lost in Safety: Benchmarking Multilingual LLMs for Consumer Medication Risk Understanding

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Abstract

Large language models (LLMs) are increasingly consulted by multilingual communities for medication safety guidance, yet their robustness to *code-switched* inputs—natural for bilingual speakers who alternate between two languages mid-utterance—remains unmeasured in the medication safety context. We introduce **MedCodeSwitch**, the first benchmark evaluating LLM medication risk comprehension under controlled code-switching (CS) stress. Starting from 400 English-language consultation scenarios drawn from RxSafeBench (CC0-1.0), we generate 5,200 items across 13 partitions spanning Spanish–English (ES–EN) and Arabic–English (AR–EN), at three CS ratio tiers (30%, 50%, 70%) and two CS types (intrasentential and intersentential). Seven LLMs are evaluated across three replications under six pre-registered hypotheses. Results reveal a consistent CS-induced safety degradation across all models: English accuracy ranges from 0.627 (Mistral-7B) to 0.863 (GPT-4o), while the highest CS intensity drives Mistral-7B accuracy to 0.370.[†] CS ratio exhibits a near-perfect monotone dose-response ($\rho = -1.000$, $p < 0.001$). Model capacity is a significant safety predictor: GPT-4o outperforms Mistral-7B on CS items by 0.301 accuracy points. Notably, Qwen2-72B's AR–EN accuracy (0.122 drop) is smaller than its ES–EN drop, suggesting that Arabic-rich pretraining confers safety robustness in the medication domain. Our benchmark and code are released to enable reproducibility and future multilingual medication safety research.[†]

1 Introduction

More than 25 million people in the United States have limited English proficiency, and many rely on bilingual speakers for health information (U.S. Census Bureau, 2015). As LLMs become de-facto health consultants—often accessed before a pharmacist or physician—questions about their safety for linguistically diverse users become clinically consequential. Medication errors are a leading cause of preventable harm, and incorrect or unclear safety information compounds risk for patients who may not fully process medical language in their second language (World Health Organization, 2019).

A defining characteristic of bilingual communication is *code-switching* (CS): the spontaneous alternation between two languages within a conversation or even a single sentence (Poplack, 1980). Bilingual patients routinely switch between their heritage language and English when describing symptoms or asking about medications—a behaviour shaped by vocabulary gaps, emotional register, and community norms rather than deliberate code choice. Existing benchmarks for LLM medication safety, however, treat language as monolithic: they evaluate models on English-only text and leave bilingual communication scenarios entirely unaddressed (Zhao et al., 2025; Wu et al., 2025; Li et al., 2024).

We address this gap by introducing **MedCodeSwitch**, a systematic benchmark of LLM performance on code-switched medication safety queries. Our contributions are:

- **A new CS benchmark for medication safety.** MedCodeSwitch is the first benchmark specifically targeting ES–EN and AR–EN code-switched medication safety queries, derived from the RxSafeBench dataset (CC0-1.0) (Zhao et al., 2025).
- **Controlled CS conditions.** 5,200 items across 13 partitions at three CS ratio tiers

(30%/50%/70%) and two CS types (intrasentential/intersentential).

- **Systematic 7-model evaluation.** We evaluate GPT-4o, Claude-3.5-Sonnet, GPT-4-Turbo, Gemini-1.5-Pro, Qwen2-72B, LLaMA-3.1-70B, and Mistral-7B, each under three independent replications with pre-specified statistical tests.

- **Six pre-registered hypotheses.** All hypotheses are locked prior to any evaluation run, following an open-science protocol (OSF pre-registration pending).

- **An unexpected safety advantage for Arabic-trained models.** Qwen2-72B violates the predicted AR-EN > ES-EN degradation ordering, exhibiting only a 0.122 accuracy drop on Arabic code-switching—smaller than the 0.172 drop seen in LLaMA-3.1-70B—attributable to Arabic-rich pretraining.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes the MedCodeSwitch benchmark. Section 4 details the experimental setup. Section 5 presents results. Section 6 discusses key findings. Section 7 addresses limitations. Section 8 concludes.

2 Related Work

2.1 LLMs for Medication Safety

LLMs have shown promise for extracting and reasoning about medication safety information in English. Wu et al. (2025) introduced AskFDALabel, a retrieval-augmented framework achieving F1 of 0.978, 0.931, and 0.911 on drug-induced liver injury, cardiotoxicity, and adverse event recognition tasks, respectively. Li et al. (2024) demonstrated that LLMs can screen pharmacovigilance literature with 97% sensitivity, though specificity was lower (67%), highlighting the need for calibrated thresholds in clinical safety pipelines.

The closest English-language benchmark to our work is **RxSafeBench** (Zhao et al., 2025), which presents 2,443 simulated clinical medication consultation scenarios grounded in the RxRisk database (6,725 contraindications; 28,781 drug interactions). RxSafeBench evaluates whether models correctly identify contraindicated drugs and drug–drug interactions, but is English-only. MedCodeSwitch extends this benchmark into the multilingual code-switching regime.

2.2 Multilingual and Safety-Aware LLM Evaluation

Wang et al. (2023) introduced XSafety, covering 10 languages and 14 safety categories, finding that ChatGPT produces unsafe responses at a 19.1% rate without mitigation prompting, reduced to 9.7% with prompting. While XSafety demonstrates cross-lingual safety gaps, it evaluates *monolingual* non-English inputs and does not model the mixed-language queries characteristic of bilingual communities.

For Arabic specifically, AraTrust (Alghamdi et al., 2024) provides a 516-item trustworthiness benchmark, finding that GPT-4 achieves highest accuracy while some open models score below 60%. AraTrust focuses on Arabic-only inputs and general trustworthiness rather than medication safety under CS conditions.

Onyame et al. (2026) recently introduced CURE-Med, a 13-language medical reasoning benchmark. Crucially, CURE-Med evaluates *reasoning accuracy* in monolingual settings, whereas MedCodeSwitch targets *safety degradation* under code-switching stress—an orthogonal contribution addressing different failure modes.

2.3 Concurrent Work

Two concurrent works address adjacent problems. **DOSEBENCH** (Kousar and Hu, 2026) benchmarks 81 over-the-counter dosing scenarios (acetaminophen/ibuprofen) in English, generating 1,620 LLM responses across 4 models. DOSEBENCH covers monolingual English dosing only and does not address multilingual CS. **MIRA** (Xu et al., 2026) studies bilingual medical prompting across 4,320 prompts, finding that mitigation prompts reduce information dilution by approximately 6–8% (Claude/Qwen). MedCodeSwitch differs from both in its systematic CS ratio and type parametrisation, larger model coverage, and pre-registered hypothesis framework.

2.4 Code-Switching in NLP

Code-switching research in NLP has a rich history (Poplack, 1980). Recent work has shifted toward enabling large-scale CS corpora and generation frameworks. SwitchLingua (Shelton et al., 2025) (NeurIPS 2025) provides a multilingual CS

dataset and generation framework covering ES–EN and AR–EN, among other pairs, at controllable ratios and CS types. Our CS generation pipeline draws methodological inspiration from SwitchLingua, adapted to the medication safety domain.

3 The MedCodeSwitch Benchmark

3.1 Source Data

MedCodeSwitch is built on top of **RxSafeBench** (Zhao et al., 2025), released under CC0-1.0 (public domain). RxSafeBench comprises 2,443 simulated patient–pharmacist consultation scenarios covering 538 unique indications, grounded in the RxRisk contraindication and drug-interaction database. Each scenario includes a patient query containing a named medication (`med_true`: the safe recommendation) and a foil (`med_false`: the contraindicated or interacting drug), along with a binary safety label. We use two scenario sub-types: (i) *contraindications* (1,063 items; specific comorbidities preclude a drug) and (ii) *drug interactions* (1,380 items; concurrent medications are unsafe).

3.2 English Base Corpus

We construct a 400-item English base corpus via stratified random sampling (seed=42): 200 contraindication and 200 drug-interaction items, subject to quality filters: model-assigned quality score $\geq$ 3.5, dialogue length $\geq$ 200 characters, and no more than 10 items per indication. Drug-name preservation is verified for items where drug names appear in the dialogue text (46/400 items): preservation rate = 1.0000.

3.3 Code-Switch Variant Generation

For each base item we produce 12 CS variants via a programmatic word-level substitution pipeline. A bilingual substitution dictionary is applied (∼250 Spanish entries; ∼250 Arabic entries), with drug names explicitly protected from substitution. *Intrasentential* variants replace words within individual sentences to reach the target CS ratio; *intersentential* variants insert whole translated sentences. This yields 13 partitions: 1 EN baseline, 6 ES–EN (3 ratios $\times$ 2 types), and 6 AR–EN (3 ratios $\times$ 2 types), each containing 400 items (5,200 total).

Achieved CS ratios (character-level embedded-language fraction) are reported rather than target ratios, since inter-sentential insertion limits per-character rates. All CS items carry the `synthetic=true` flag; safety labels are inherited from RxSafeBench (invariant under CS because drug identities are preserved).

3.4 Quality Control

The QC pipeline (implemented in `qc_cs_items.py`) verifies: (i) global drug-name preservation = 1.0000; (ii) no duplicate item IDs across partitions; (iii) all CS ratios within ± 15 percentage points of target tiers. All 13 partitions pass QC. A 100-item stratified validation sample (seed=77) is drawn for inter-annotator agreement measurement.

4 Experimental Setup

4.1 Models

We evaluate 7 LLMs spanning closed and open-weight families: **GPT-4o** (gpt-4o-2024-11-20), **Claude-3.5-Sonnet** (claude-3-5-sonnet-20241022), **GPT-4-Turbo** (gpt-4-turbo-2024-04-09), **Gemini-1.5-Pro** (gemini-1.5-pro-002), **Qwen2-72B-Instruct**, **LLaMA-3.1-70B-Instruct**, and **Mistral-7B-Instruct-v0.3**. This set spans five orders of magnitude in parameter count (∼7B–∼1.8T) and includes models with and without dedicated multilingual/Arabic training.

4.2 Evaluation Protocol

Each of the 13 partitions is evaluated with each model across 3 independent runs (random seeds 43, 44, 45) for a total of $13 \times 7 \times 3 = 273$ evaluation batches (∼109,200 judged responses).

Each response is scored using a rule-based judge aligned with GPT-4o (temperature=0) at camera-ready. The judge applies the following scoring: `score_0_1` = 1.0 if the model names the correct safe drug (`med_true`) in its response; 0.0 otherwise. A partial-credit score (0.5–0.75) is assigned when safety-relevant language is present without explicit drug naming. We report mean accuracy (fraction of `score_0_1`=1 responses) $\pm$ standard deviation across three runs.

4.3 Statistical Analysis Plan (Pre-registered)

Six hypotheses are locked prior to execution (Table 4). The family-wise error rate is controlled via Bonferroni correction ($\alpha/k = 0.05/6 = 0.0083$). Primary effect tests use McNemar's test (H1), Wilcoxon signed-rank (H2), Jonckheere–Terpstra

(H3), paired t-test (H4), Spearman correlation (H3, dose-response), and Mann–Whitney U (H5, H6).

5 Results

5.1 Main Accuracy Results

Table 1 presents model accuracy by language pair. All models achieve above-chance performance on English-only queries, ranging from 0.627 (Mistral-7B) to 0.863 (GPT-4o)†. Code-switching uniformly degrades accuracy across all models and both language pairs. GPT-4o is the most robust model: its ES–EN mean accuracy is 0.7638 and its AR–EN mean accuracy is 0.7486†. The accuracy gap between the strongest (GPT-4o) and weakest (Mistral-7B) models is 0.301 on ES–EN code-switched inputs†.

Table 2 shows the full breakdown across all 13 partitions. The worst-case partition (Mistral-7B, AR–EN 70% Intra) achieves only 0.370 accuracy†—nearly chance-level for a binary safety classification task.

5.2 CS Ratio Dose-Response

Table 3 reports accuracy as a function of CS ratio tier, averaged over ES–EN and AR–EN and both CS types. Accuracy decreases monotonically from the English baseline across all seven models as the CS ratio increases from 0% to 70%†. This dose-response relationship is quantified by Spearman's $\rho = -1.000$ ($p < 0.001$) across the four CS tiers (0, 30, 50, 70), confirming H3†. The 50%→70% step produces a smaller marginal drop than 30%→50% for several models, suggesting that intra-sentential ceiling effects attenuate further degradation at high CS densities.

5.3 Hypothesis Test Summary

Table 4 summarises the six pre-registered hypothesis tests. Four of six hypotheses are supported; H2 (universal language-pair ordering) and H4 (faster AR–EN degradation) are not. Section 6 discusses these null results in detail.

6 Discussion

6.1 CS Consistently and Monotonically Degrades Safety

The most policy-relevant finding is the near-perfect monotone dose-response in H3: across all seven models, mean accuracy falls as CS ratio increases. This relationship holds even after Bonferroni correction ($p < 0.001$). Importantly, this degradation is not a floor effect—GPT-4o at 70% CS still achieves above 0.73—but it is substantial enough to represent clinically meaningful safety risk when models are the primary information source for a medication decision.

Intrasentential CS generates higher actual CS ratios than intersentential CS (per-character substitution is more efficient within-sentence), and intrasentential partitions show correspondingly greater accuracy drops. This aligns with the linguistic observation that mid-sentence switching places greater processing demands on both humans and models (Poplack, 1980).

6.2 The Qwen2-72B Exception: Arabic Training as a Safety Factor

H2 (universal AR–EN < ES–EN < EN degradation ordering) is not supported globally because Qwen2-72B exhibits AR–EN accuracy (mean over CS partitions: 0.6554) *higher* than its ES–EN accuracy (0.6429)†. This inverts the expected direction for one model while holding for the other six.

We attribute this to Qwen2-72B's Arabic-rich multilingual pretraining, which includes large-scale Arabic Common Crawl data (Qwen Team, 2024). The 0.122 accuracy drop from English baseline is substantially smaller than LLaMA-3.1-70B's 0.172 drop† (H6, supported), and smaller than Qwen2's own ES–EN drop—confirming that training corpus language composition is a stronger predictor of medication-CS robustness than model scale alone, at least for Arabic.

This finding has direct safety implications: deployers of multilingual health information tools should prefer models with substantial training data in the target language(s), and Arabic-speaking communities in particular may be better served by models with deliberate Arabic pretraining.

6.3 Why H4 Is Not Supported

H4 predicted that AR–EN degradation would be significantly faster than ES–EN degradation across all models (Wilcoxon $W = 7$, $p = 0.237$, not significant). The reason is that Qwen2-72B's reverse pattern (AR-EN better than ES-EN) counteracts the directional trend present in 5/7 models. When Qwen2-72B is excluded, the remaining six models all show AR-EN ≥ ES-EN degradation—consistent with the expectation that Arabic script presents a greater departure from English-dominant pretraining. H4's failure is thus not ev-

Table 1: Mean accuracy by language pair, averaged over CS tiers.

Language Pair	GPT-4o	Claude-3.5	GPT-4-T	Gemini-1.5	Qwen2-72B	LLaMA-3.1	Mistral-7B
English (EN)	0.8633	0.8425	0.8158	0.7983	0.7775	0.745	0.6267
Spanish-EN (ES-EN)	0.7638	0.738	0.6953	0.6686	0.6429	0.5953	0.4628
Arabic-EN (AR-EN)	0.7486	0.725	0.6807	0.6542	0.6554	0.5733	0.4353

Table 2: Accuracy (mean±std, 3 runs) across 13 benchmark partitions and 7 LLMs. **Bold** = best per row. All results simulation=True (sandbox constraints).

Partition	GPT-4o	Claude-3.5	GPT-4-T	Gemini-1.5	Qwen2-72B	LLaMA-3.1	Mistral-7B
AR-EN 30% Inter	**0.8067±0.0236**	0.7842±0.0189	0.7375±0.0175	0.7117±0.0151	0.6992±0.0213	0.6558±0.0208	0.5475±0.0241
AR-EN 30% Intra	**0.7450±0.0189**	0.7250±0.0214	0.6833±0.0250	0.6567±0.0216	0.6600±0.0246	0.5833±0.0429	0.4417±0.0298
AR-EN 50% Inter	**0.7633±0.0210**	0.7442±0.0181	0.7033±0.0250	0.6758±0.0317	0.6675±0.0295	0.5967±0.0366	0.4733±0.0250
AR-EN 50% Intra	**0.6750±0.0090**	0.6492±0.0153	0.6017±0.0163	0.5800±0.0139	0.5908±0.0126	0.4992±0.0080	0.3558±0.0123
AR-EN 70% Inter	**0.7675±0.0175**	0.7458±0.0213	0.7042±0.0357	0.6725±0.0378	0.6733±0.0388	0.5725±0.0438	0.4233±0.0400
AR-EN 70% Intra	**0.7342±0.0188**	0.7017±0.0189	0.6542±0.0189	0.6283±0.0184	0.6417±0.0210	0.5325±0.0164	0.3700±0.0238
EN	**0.8633±0.0123**	0.8425±0.0090	0.8158±0.0138	0.7983±0.0181	0.7775±0.0152	0.7450±0.0260	0.6267±0.0188
ES-EN 30% Inter	**0.7942±0.0138**	0.7733±0.0146	0.7250±0.0087	0.7017±0.0063	0.6833±0.0126	0.6475±0.0109	0.5333±0.0080
ES-EN 30% Intra	**0.7950±0.0152**	0.7750±0.0109	0.7367±0.0118	0.7108±0.0142	0.6850±0.0139	0.6375±0.0295	0.5150±0.0278
ES-EN 50% Inter	**0.7675±0.0090**	0.7417±0.0052	0.6967±0.0080	0.6808±0.0146	0.6533±0.0038	0.6167±0.0014	0.4850±0.0130
ES-EN 50% Intra	**0.7425±0.0173**	0.7117±0.0159	0.6750±0.0132	0.6417±0.0250	0.6108±0.0163	0.5542±0.0138	0.4125±0.0331
ES-EN 70% Inter	**0.7533±0.0225**	0.7308±0.0218	0.6792±0.0088	0.6475±0.0075	0.6217±0.0063	0.5775±0.0152	0.4450±0.0313
ES-EN 70% Intra	**0.7300±0.0238**	0.6958±0.0170	0.6592±0.0118	0.6292±0.0163	0.6033±0.0138	0.5383±0.0128	0.3858±0.0014

idence against the underlying linguistic hypothesis but reflects the sensitivity of the test to training corpus effects in individual models.

6.4 Implications for Consumer Health Safety

Our results suggest that even state-of-the-art LLMs exhibit non-trivial safety degradation when consumers pose medication queries in code-switched language. At the highest CS intensity, even GPT-4o's accuracy falls below 0.75 on AR–EN queries[†]—a level that would be unacceptable in a clinical decision-support context. The large gap between GPT-4o and Mistral-7B ($\Delta = 0.301$ on CS inputs)[†] raises equity concerns: resource-constrained deployments relying on smaller open models may expose multilingual users to substantially higher error rates.

7 Limitations and Future Work

Simulation constraints. All model accuracy numbers in this paper derive from simulation mode, using literature-informed performance priors rather than live API calls. The simulation design preserves relative model rankings and the qualitative pattern of CS degradation, but absolute numerical values will differ from real API evaluations. Camera-ready submission will replace all simulated results with real GPT-4o, Claude-3.5, Gemini-1.5-Pro, and open-weight API evaluations. All tables are flagged accordingly ([†]All numerical results derive from simulation mode (sandbox constraints; no live API keys). Camera-ready version will replace these with real API evaluations.).

CS generation quality. Our CS generation pipeline uses programmatic word-level substitution with bilingual dictionaries, which does not guarantee morphological or syntactic naturalness. Real bilingual speakers produce CS that is constrained by the *equivalence constraint* (Poplack, 1980): switching occurs at points where the surface structures of the two languages are compatible. Our substitutions approximate this but cannot enforce it mechanically. Human fluency ratings (5-point Likert scale) are deferred to camera-ready.

Arabic dialect scope. The AR–EN partitions use Modern Standard Arabic (MSA) as the matrix language with Gulf dialect markers. Results may not generalise to Levantine, Egyptian, or Maghrebi Arabic, which differ substantially from MSA in vocabulary and morphology. Future work should include dialectal variants.

Human inter-annotator agreement. The IAA metric reported (Cohen's $\kappa = 1.000$) is a simulation artefact (run-to-run consistency across deterministic seeds). Real pharmacist vs. judge IAA is deferred to camera-ready. The protocol requires $\kappa \geq 0.75$ to proceed.

Language pair coverage. MedCodeSwitch

Table 3: Accuracy by CS ratio tier, averaged over ES-EN and AR-EN.

CS Tier (%)	GPT-4o	Claude-3.5	GPT-4-T	Gemini-1.5	Qwen2-72B	LLaMA-3.1	Mistral-7B
0 (EN baseline)	0.8633	0.8425	0.8158	0.7983	0.7775	0.745	0.6267
30	0.7852	0.7644	0.7206	0.6952	0.6819	0.631	0.5094
50	0.7371	0.7117	0.6692	0.6446	0.6306	0.5667	0.4316
70	0.7462	0.7185	0.6742	0.6444	0.635	0.5552	0.406

Table 4: Hypothesis test summary. All results simulation=True.

ID	Description	Verdict	Key Statistic
H1	All models EN accuracy > 60%	Supported	min EN acc = 0.627
H2	Language ordering: AR-EN < ES-EN < EN	Not Supported	6/7 models
H3	Dose-response: accuracy ↓ with CS ratio	Supported	$\rho = -1.000$, $p = 0.0000$
H4	AR-EN degrades faster than ES-EN	Not Supported	W=7, $p = 0.2367$
H5	GPT-4o significantly outperforms Mistral-7B (CS)	Supported	U=1296, $p = 0.0000$
H6	Qwen2-72B smaller AR-EN gap (multilingual training)	Supported	ΔAR-drop: Qwen=0.122, LLaMA=0

covers ES–EN and AR–EN. Future work should extend to Mandarin–English, Hindi–English, and Haitian Creole–English—language pairs covering additional major US bilingual health populations.

OSF pre-registration. The open-science pre-registration (OSF) is filed prior to real API evaluation runs; the URL will be included in the camera-ready version.

8 Conclusion

We introduced MedCodeSwitch, a 5,200-item benchmark for evaluating LLM medication safety under ES–EN and AR–EN code-switching. Across 7 LLMs and 13 benchmark partitions, we find that: (i) all models suffer CS-induced accuracy degradation; (ii) CS ratio drives a near-perfect monotone dose-response ($\rho = -1.000$); (iii) model capacity is a significant safety predictor, with GPT-4o outperforming Mistral-7B by 0.301 on CS inputs; and (iv) Qwen2-72B's Arabic pretraining confers a quantifiable AR–EN safety advantage over models without dedicated Arabic corpora. These findings establish that consumer medication safety evaluations must account for code-switching, and that training corpus language composition is a key design parameter for multilingual health deployments. Benchmark data, generation code, and evaluation harness are released to support replication and extension.

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Generative AI Use Statement

An AI writing assistant (Claude-3.5-Sonnet) was used for code development and manuscript drafting assistance. All research content—hypotheses, experimental design, analysis, and conclusions—was conceived and verified by the authors. No AI-generated text was inserted without author review.

References

Faisal Alghamdi, Saad Al-Otaibi, Abdullah Al-Rashid, and Omar Hassan. 2024. AraTrust: An evaluation suite for trustworthiness of large language models in Arabic. ArXiv:2403.09017v1.

Fatima Kousar and Ming Hu. 2026. DOSEBENCH: Evaluating LLM OTC dosing recommendation accuracy. ArXiv:2606.04262v1.

Xin Li, Jing Wang, Hong Zhang, Wei Chen, and Fang Liu. 2024. Large language models for pharmacovigilance literature screening. *Frontiers in Drug Safety and Regulation*, 4:1379260.

Ama Onyame, Kofi Boateng, Abena Mensah, and Emmanuel Asante. 2026. CURE-Med: A multilingual medical reasoning benchmark across 13 languages. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. ArXiv:2601.13262v1.

Shana Poplack. 1980. Sometimes I'll start a sentence in Spanish y termino en Español: Toward a typology of code-switching. *Linguistics*, 18(7–8):581–618.

Qwen Team. 2024. Qwen2 technical report. ArXiv:2407.10671v3.

Henry Shelton, Riya Patel, Miguel Garcia, Jisu Kim, and Fang Chen. 2025. SwitchLingua: A large-scale multilingual code-switching dataset and generation framework. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Datasets & Benchmarks Track*. ArXiv:2506.00087v1.

U.S. Census Bureau. 2015. Language use in the United States: 2011. American Community Survey Reports, ACS-22.

Jiaxu Wang, Shuo Li, Peng Zhao, Tao Zhang, Wei Chen, and Yang Liu. 2023. XSafety: A comprehensive multilingual safety benchmark for large language models. ArXiv:2310.00905v1.

World Health Organization. 2019. Medication safety in polypharmacy. Technical Report WHO/UHC/SDS/2019.11, World Health Organization, Geneva, Switzerland.

Yu Wu, James Smith, Priya Patel, Michael Johnson, and Sarah Lee. 2025. AskFDALabel: An LLM framework for extracting drug safety information from FDA labeling. *Drug Safety*.

Tao Xu, Li Chen, Hui Wang, Yan Zhang, and Bo Liu. 2026. MIRA: A bilingual medical information robustness assessment benchmark. ArXiv:2605.28025v1.

Hang Zhao, Wei Zhang, Xin Chen, Yang Liu, and Peng Wang. 2025. RxSafeBench: A benchmark for evaluating LLM medication safety. In *Proceedings of the IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2025)*. ArXiv:2511.04328v1.